

KALAOA KONA, HI, USA

WMO: 911975

Lat: 19.736N

Lon: 156.049W

Elev: 43

StdP: 14.67

Time Zone: -10.00 (W10)

Period: 97-19

WBAN: 21510

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	66.0	67.6	51.4	56.3	72.2	53.8	61.7	72.3	29.6	78.1	26.2	78.8	6.0	70	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	88.1	76.2	87.0	75.4	86.1	74.8	77.8	85.6	77.0	84.6	76.3	83.9	12.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	132.7	83.2	74.4	128.6	82.7	73.4	124.2	82.1	41.2	85.5	40.4	84.8	39.7	83.8	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	16.9	14.8	DB	62.5	90.6	1.7	3.5	61.3	93.2	60.3	95.2	59.3	97.2	58.0	99.8	(4)
(5)			WB	54.7	78.2	2.9	1.9	52.6	79.5	50.9	80.6	49.3	81.7	47.3	83.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.9	74.8	74.7	75.6	77.0	78.0	79.1	80.1	81.0	80.6	79.9	77.9	75.9
	DBStd	2.75	1.63	1.80	1.76	1.61	1.53	1.49	1.73	1.85	1.77	1.70	1.60	1.61
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10184	769	692	795	809	867	873	933	961	919	926	838	802
	CDD65	4709	304	272	330	359	402	423	468	496	469	461	388	337
	CDH74	39311	1690	1491	1953	2492	3202	3882	4693	5236	4892	4536	3138	2106
	CDH80	7865	99	93	172	305	484	769	1153	1496	1383	1161	549	201
Wind	WSAvg	7.7	7.7	8.0	8.0	8.0	7.8	7.8	7.9	7.9	7.7	7.4	7.2	7.4
Precipitation	PrecAvg	28.1	2.6	1.5	2.2	2.1	2.1	2.7	2.8	2.8	2.9	2.5	2.0	1.9
	PrecMax	54.4	7.2	3.7	6.1	10.5	5.0	5.6	7.0	6.1	5.9	6.1	6.1	7.7
	PrecMin	16.8	0.0	0.0	0.4	0.9	0.4	0.6	0.9	1.5	0.8	0.3	0.3	0.0
	PrecStd	9.2	2.0	1.1	1.5	1.9	1.1	1.3	1.4	1.1	1.2	1.5	1.6	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.2	83.5	84.2	84.7	85.5	86.9	88.2	89.6	89.4	88.4	86.1	84.2
		MCWB	69.2	71.2	70.0	71.7	73.4	74.0	76.2	77.5	77.3	76.4	73.1	70.4
	2%	DB	81.9	82.0	82.5	83.7	84.3	85.4	86.6	88.0	88.0	86.9	84.5	82.5
		MCWB	70.0	70.7	71.0	72.0	73.0	73.6	75.3	76.4	76.4	75.2	73.0	71.0
	5%	DB	81.2	81.2	81.8	82.6	83.5	84.4	85.6	86.5	86.4	85.8	83.8	81.9
		MCWB	70.1	70.3	70.9	71.5	72.8	73.3	74.8	75.5	75.4	74.6	72.9	70.8
	10%	DB	80.5	80.5	81.0	81.8	82.4	83.7	84.4	85.6	85.5	84.5	82.7	81.2
		MCWB	69.7	69.7	70.6	71.0	72.2	73.1	74.2	75.0	74.7	74.1	72.5	70.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.4	74.4	75.2	75.0	76.1	76.3	77.6	78.7	79.0	78.2	76.3	75.1
		MCDB	80.3	80.6	81.0	82.0	82.4	84.0	85.4	86.9	87.0	85.9	83.1	80.7
	2%	WB	73.2	73.2	73.7	73.7	74.9	75.4	76.7	77.7	77.8	76.9	75.3	73.5
		MCDB	79.4	80.3	80.1	81.1	81.8	82.7	84.5	85.5	85.5	84.4	82.3	80.0
	5%	WB	72.0	72.1	72.7	73.0	74.0	74.8	76.0	76.9	76.8	76.0	74.6	72.9
		MCDB	79.0	79.3	79.7	80.8	81.3	82.4	83.7	84.6	84.5	83.6	81.8	79.7
	10%	WB	71.0	71.0	71.5	72.2	73.3	73.9	75.3	76.1	75.9	75.2	73.5	72.0
		MCDB	78.5	78.4	79.3	80.3	80.9	81.9	83.0	83.8	83.7	83.0	81.1	79.3
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.8	11.2	10.6	10.2	10.0	10.1	10.4	10.7	10.7	10.9	11.4
		MCDBR	12.2	12.2	11.9	11.2	10.8	10.7	10.7	11.0	11.3	11.4	11.5	12.0
		MCWBR	7.6	7.5	7.1	6.6	6.2	6.4	6.3	6.5	6.8	6.9	7.0	7.5
	5% WB	MCDBR	11.3	11.0	10.9	10.7	10.2	10.2	10.2	10.3	10.6	10.6	10.5	10.7
	MCWBR	7.6	7.3	7.0	6.6	6.1	6.4	6.0	6.3	6.7	6.8	6.9	7.1	
Clear-Sky Solar Irradiance	taub	0.361	0.367	0.383	0.423	0.412	0.390	0.399	0.385	0.375	0.379	0.375	0.357	
	taud	2.582	2.565	2.518	2.374	2.443	2.508	2.486	2.536	2.569	2.575	2.581	2.614	
	Ebn at Noon	292	296	293	281	280	284	282	288	292	288	285	290	
	Edh at Noon	30	31	34	40	37	34	35	33	32	31	29	28	
All-Sky Solar Radiation	RadAvg	1467	1610	1753	1949	2029	2087	2117	2072	1924	1679	1447	1337	
	RadStd	74	102	178	126	120	68	71	91	64	61	61	118	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+225

Nomenclature: See separate page